A Product and Service Management Consulting Framework for Sri Lankan Tech Companies

Marketing Strategy Assessment

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan technology companies increasingly operate in a digital environment where effective marketing is important for reaching and communicating with potential customers. However, technology products and services often contain technical information that may not be equally understandable to customers with different levels of technical knowledge. In addition, selecting appropriate digital marketing channels and allocating limited marketing budgets can be challenging for companies seeking strategies suited to the Sri Lankan market. Previous Sri Lankan research has examined digital marketing adoption among SMEs and identified factors influencing the use of digital marketing strategies, while research on Sri Lankan technology startups has also examined challenges affecting startup success.

However, there is a gap in integrated marketing decision-support solutions specifically designed for Sri Lankan technology companies that combine audience technical literacy analysis, technical feature-to-value translation, marketing channel recommendation, and budget allocation. Therefore, this research proposes a **Marketing Strategy Assessment**, an AI-based marketing decision-support component that analyzes target audience information, generates simplified customer-oriented product messages, ranks suitable marketing channels, and recommends marketing budget allocation.

The research will use a mixed-method approach involving literature review, questionnaire/interview-based primary data collection, and experimental evaluation. Natural Language Processing and AI techniques will be used for audience and text analysis, while the Analytical Hierarchy Process (AHP) will be used to rank marketing channels. The system will be developed through an iterative prototyping approach and evaluated using classification performance, expert assessment of generated marketing messages, recommendation relevance, and usability measures. The component will provide marketing intelligence and recommendations that can be integrated with the other components of the overall consulting framework.

Keywords: Marketing Strategy, Digital Marketing, Technical Literacy, Marketing Recommendation, Sri Lankan Technology Companies

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LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligence
AHP	Analytical Hierarchy Process
API	Application Programming Interface
B2B	Business-to-Business
B2C	Business-to-Consumer
DigComp	Digital Competence Framework
ISE	Information System Engineering
LKR	Sri Lankan Rupee
ML	Machine Learning
NLP	Natural Language Processing
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SME	Small and Medium-sized Enterprise
SUS	System Usability Scale
TOE	Technology–Organization–Environment
WBS	Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

The rapid growth of digital technologies has changed the way businesses communicate with customers, promote products and services, and conduct marketing activities. Digital marketing provides businesses with opportunities to reach customers through online platforms and to use digital information when planning marketing activities. In the Sri Lankan context, research has examined the adoption of digital marketing among small and medium-sized enterprises (SMEs). Madurapperuma and Colombage investigated factors influencing digital marketing adoption by Sri Lankan SMEs, with particular attention to social media marketing, and used the Technology–Organization–Environment (TOE) framework to structure the study. Their proposed research strategy combined online surveys and interviews, demonstrating the relevance of examining digital marketing practices within the Sri Lankan business environment.

The importance of digital marketing is further supported by the increasing use of internet and social media services in Sri Lanka. According to DataReportal's Digital 2026 report, there were approximately 13.9 million internet users in Sri Lanka at the end of 2025, representing an internet penetration rate of 59.7%. The report also identified approximately 9.0 million social media user identities, equivalent to 38.7% of the country's population. These figures demonstrate the importance of digital channels as potential communication and marketing environments for businesses operating in Sri Lanka.

Technology companies face additional challenges when marketing products and services because their offerings may contain technical features, terminology, and concepts that are not equally understandable to all potential customers. A technology product may therefore need to be communicated differently depending on the technical knowledge and information needs of its target audience. For a technology company, simply describing technical features may not effectively communicate the practical value of a product to customers. This creates a need for marketing approaches that can consider the characteristics of the target audience and translate technical product information into understandable customer-oriented value.

The selection of suitable marketing channels and allocation of marketing budgets also require consideration of the target audience and campaign objectives. Although digital platforms provide businesses with multiple channels for reaching customers, the suitability of a particular channel can vary according to factors such as audience characteristics, campaign objectives, expected reach, cost, and engagement potential. Sri Lankan technology companies may therefore benefit from a decision-support approach that considers local market information rather than relying only on generalized global marketing recommendations.

Existing digital marketing platforms provide functions such as advertising, audience targeting, campaign management, and performance measurement. However, these functions do not necessarily provide an integrated process that combines audience technical-literacy analysis, technical feature-to-value translation, local marketing-channel recommendation, and budget allocation specifically for Sri Lankan technology companies. Research on Sri Lankan SMEs has examined digital marketing adoption, while research on Sri Lankan technology startups has investigated broader factors associated with startup failure and the need for

informed entrepreneurial decision-making. However, these studies do not establish an integrated AI-based marketing decision-support solution addressing the combination of needs proposed in this research.

To address this problem, this research proposes a **Marketing Strategy Assessment**, which is an AI-based marketing decision-support component for Sri Lankan technology companies. The proposed system will analyze target audience information to estimate audience technical literacy, assist in converting technical product features into customer-oriented value statements, rank suitable marketing channels using an Analytical Hierarchy Process (AHP)-based approach, and recommend the allocation of an available marketing budget. Sri Lankan digital-market information will be incorporated to provide context for marketing recommendations. The expected outputs include an audience profile and technical literacy level, simplified marketing messages, a ranked channel suitability matrix, and a recommended campaign budget plan.

The proposed component supports the broader research project by providing the marketing intelligence and recommendations required by the overall Product and Service Management Consulting Framework. It is also expected to contribute to relevant Sustainable Development Goals (SDGs), particularly SDG 8, Decent Work and Economic Growth, through supporting technology businesses and entrepreneurship, and SDG 9, Industry, Innovation and Infrastructure, through the application of AI-based digital technology to business decision-making. The final SDG alignment will be refined according to the research cluster and institutional requirements of the project.

1.2 Literature Review

R. K. Madurapperuma and V. K. Colombage, "Factors Affecting the Adoption of Digital Marketing by SMEs in the Context of Sri Lanka," 2022.

Madurapperuma and Colombage [1] investigated the factors affecting the adoption of digital marketing by small and medium-sized enterprises (SMEs) in Sri Lanka. The study examined digital marketing adoption within the Sri Lankan context using the Technology–Organization–Environment (TOE) framework. The research considered technological, organizational, and environmental factors that may influence the adoption of digital marketing practices. The study is relevant to the proposed Smart Marketing System because it provides evidence that digital marketing should be examined according to the characteristics of the Sri Lankan business environment. However, the study focuses on factors influencing digital marketing adoption rather than providing an AI-based system for audience technical literacy analysis, marketing message simplification, channel recommendation, or budget allocation. Therefore, the study provides contextual support for the proposed research while leaving opportunities for an integrated marketing decision-support solution.

K. Wimal and S. Ajendra, “A Phenomenological Exploration of Technology Start-Up Failure in Sri Lanka,” Journal of Entrepreneurship, Management, and Innovation, vol. 19, no. 3, pp. 47–86, 2023.

Wimal and Ajendra [2] conducted a phenomenological study to explore technology start-up failure in Sri Lanka. The study examined the experiences of technology start-up leaders and investigated factors associated with start-up failure within the Sri Lankan context. The study is relevant to the proposed Smart Marketing System because it provides context regarding the challenges faced by technology start-ups operating in Sri Lanka. However, the study focuses on understanding technology start-up failure rather than developing a marketing decision-support solution. It does not specifically address audience technical literacy, technical feature-to-value translation, marketing channel selection, or marketing budget allocation. Therefore, the study provides contextual evidence for the challenges of Sri Lankan technology start-ups while leaving room for research into specialized marketing decision-support approaches.

European Commission's DigComp 2.2 Digital Competence Framework.

The Digital Competence Framework for Citizens (DigComp) provides a structured framework for understanding digital competence through knowledge, skills, and attitudes related to the effective and responsible use of digital technologies. The framework provides a useful conceptual basis for considering differences in users' levels of digital competence when designing technology-related communication. However, DigComp is a general digital competence framework rather than a marketing-oriented model for determining the level of technical detail that should be used when communicating a technology product to a specific target audience. Therefore, the framework can inform the conceptualization of audience technical literacy in the proposed Smart Marketing System, while the proposed research will investigate how audience information and textual inputs can be used to derive a marketing-oriented technical literacy classification.

Li, Q, Peng, H, Li, J et al. (5 more authors) (2022) *A Survey on Text Classification: From Traditional to Deep Learning.* ACM Transactions on Intelligent Systems and Technology, 13 (2). 31. ISSN: 2157-6904

(Natural Language Processing and Text Classification)

Li et al. [4] reviewed text classification techniques from traditional machine learning approaches to deep learning methods and identified text classification as an important task in Natural Language Processing (NLP). The study discusses different approaches to text representation, feature extraction, classification models, datasets, and evaluation metrics used for text classification. The review demonstrates that textual information can be processed and assigned to predefined categories using different machine learning and deep learning approaches. A strength of the study is its broad comparison of classification approaches and evaluation considerations. However, the study provides a general review of text classification rather than addressing the classification of target audiences according to their technical literacy for technology-product marketing. Therefore, its findings provide a technical foundation for investigating an NLP-assisted audience technical literacy classifier within the proposed Smart Marketing System.

Feature-to-Value Translation and Customer Value

The Value Proposition Canvas provides a framework for understanding the relationship between products or services and customer needs by considering customer jobs, pains, gains, and the corresponding products, pain relievers, and gain creators. This framework is relevant to the proposed Smart Marketing System because it provides a conceptual basis for connecting product features with the value that those features can provide to customers. However, the framework itself is primarily a business and value-design tool and does not provide an automated method for transforming technical product descriptions into customer-oriented marketing messages. Therefore, the proposed research will investigate the use of AI and Natural Language Processing techniques to assist in translating technical product features into simpler value-oriented statements while considering the characteristics of the target audience.

E. W. T. Ngai, “Selection of web sites for online advertising using the AHP,” Information & Management, vol. 40, no. 4, pp. 233–242, 2003.

(Marketing Channel Selection Using Analytical Hierarchy Process)

Ngai [5] investigated the selection of websites for online advertising using the Analytical Hierarchy Process (AHP), a multi-criteria decision-making technique. The study considered multiple criteria when evaluating alternative advertising websites, demonstrating the applicability of AHP to online advertising channel selection. This approach is relevant to the proposed Smart Marketing System because marketing-channel selection may depend on several factors rather than a single criterion. AHP can provide a structured method for comparing alternative channels according to criteria such as audience suitability, cost, reach, and other marketing considerations. However, the study focuses on website selection for online advertising and does not specifically address Sri Lankan technology companies or integrate audience technical literacy, local market information, and marketing budget allocation. Therefore, the

proposed research will investigate an AHP-based channel-ranking approach adapted to the requirements of Sri Lankan technology-company marketing.

Digital Marketing Budget Allocation

Marketing budget allocation is an important decision in digital marketing because organizations generally have limited financial resources that need to be distributed among different marketing activities and channels. The allocation decision can involve multiple factors, including campaign objectives, target audience, channel suitability, expected outcomes, and available budget. Existing digital advertising platforms provide budgeting and campaign-management functions, but these tools are primarily designed for executing campaigns within their respective advertising ecosystems. They do not necessarily provide an integrated budget allocation approach that combines audience technical literacy, channel suitability, and Sri Lankan market context across multiple marketing channels. Therefore, the proposed Smart Marketing System will investigate a decision-support approach for recommending the allocation of a user-defined marketing budget among suitable channels. The allocation approach will be informed by the channel-ranking results and relevant evidence collected through the research.

Critical Analysis of Existing Research

Ref.	Research Area	Method / Approach	Context / Data	Main Contribution	Limitation Relevant to This Research
[1]	Digital marketing adoption	TOE-based study	Sri Lankan SMEs	Identifies factors influencing digital marketing adoption	Does not develop an integrated AI marketing decision-support system
[2]	Technology startup challenges	Phenomenological study	Sri Lankan technology startups	Provides insight into challenges experienced by technology startups	Does not specifically address marketing decision support
[3]	Digital competence	Competence framework	General digital competence	Provides a conceptual basis for understanding digital competence	Not designed for marketing-oriented technical literacy classification
[4]	NLP and text classification	NLP / machine-learning review	Text-classification research	Provides technical foundation for text classification	Does not classify audience technical literacy for technology marketing
[5]	Marketing channel selection	AHP	Online advertising	Demonstrates AHP for multi-criteria advertising selection	Not tailored to Sri Lankan technology companies or integrated with technical literacy
[6]	Marketing budget allocation	To be finalized	To be finalized	To be finalized	To be finalized

Table 1.1.Critical Analysis of Existing

1.3 Research Gap

Existing research has examined digital marketing adoption in Sri Lanka, challenges faced by technology start ups, digital competence, NLP based text classification, customer value communication, and multi criteria approaches for advertising channel selection. However, these areas are addressed separately, and there is limited evidence of an integrated AI based marketing decision support solution specifically designed for Sri Lankan technology companies. In particular, existing approaches do not combine audience technical literacy analysis, technical feature to value translation, Sri Lankan context marketing channel recommendation, and marketing budget allocation within a single system. Therefore, this research investigates the development and evaluation of a Marketing Strategy Assessment that integrates these functions to support marketing decision making for Sri Lankan technology companies.

Specific Research Gap

“The Marketing Strategy Assessment is an AI-based marketing decision-support component designed for Sri Lankan technology companies. It analyzes target-audience technical literacy, translates technical product features into customer-oriented value statements, recommends suitable marketing channels using multi-criteria analysis, and allocates marketing budgets based on campaign requirements and available resources.”

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate an AI based Marketing Strategy Assessment that analyzes audience technical literacy, translates technical product features into customer oriented messages, and recommends suitable marketing channels and budget allocation for Sri Lankan technology companies.

2.2 Specific Objectives

1. To identify the marketing communication, audience analysis, channel selection, and budget allocation requirements of Sri Lankan technology companies through literature review and primary data collection.
2. To develop an NLP assisted approach for classifying target audiences according to their technical literacy levels using structured questionnaire responses and textual audience information.
3. To develop an AI based feature to value translation module that converts technical product features into clear and customer-oriented marketing messages based on target audience characteristics.
4. To develop an AHP based marketing channel recommendation model that ranks suitable marketing channels according to audience characteristics, campaign objectives, and relevant market criteria.
5. To develop and evaluate a marketing budget allocation approach that distributes a user defined budget among recommended channels and assess the effectiveness and usability of the proposed Smart Marketing System.

3. METHODOLOGY

3.1 Research Methodology

This research will adopt a mixed methods research approach combined with an experimental and system development oriented design. The mixed method approach is appropriate because the research requires both quantitative and qualitative information to understand the marketing requirements of Sri Lankan technology companies and their target audiences. Quantitative data will be collected through structured questionnaire responses to identify audience characteristics, technical familiarity, marketing channel preferences, campaign objectives, and budget related information. Qualitative information will be collected through open ended responses and, where feasible, interviews with relevant stakeholders to obtain deeper insights into marketing communication and decision making challenges.

The collected data will be used to inform the design of the proposed Marketing Strategy Assessment**.** The system will incorporate NLP assisted audience technical literacy analysis, AI based feature to value translation, AHP based marketing channel recommendation, and marketing budget allocation. The developed system will then be evaluated using appropriate performance, expert assessment, recommendation, and usability measures. This approach is suitable because the research not only investigates a practical marketing problem but also develops and evaluates a software-based decision support solution.

3.2 Data / Participants / Experimental Inputs

The study will use both primary and secondary data to support the development and evaluation of the Smart Marketing System. Primary data will be collected from relevant Sri Lankan technology-company stakeholders, including founders, owners, managers, and marketing professionals involved in promoting technology products or services. The primary data will capture company and product characteristics, target audience characteristics, audience technical familiarity, marketing communication preferences, currently used marketing channels, campaign objectives, and marketing budget information.

Data will be collected using a structured questionnaire containing closed-ended and open-ended questions. Where feasible, semi-structured interviews will also be conducted with selected stakeholders to obtain deeper insights into marketing communication, channel selection, and budget allocation challenges. The final sample size and sampling strategy will be determined based on participant accessibility and appropriate research sampling considerations.

Secondary data will be collected from credible Sri Lankan sources, including academic publications, industry reports, and digital-market statistics. These data will be used to develop a Sri Lankan marketing knowledge base containing relevant information about digital-market characteristics and marketing channels.

Experimental inputs will consist of representative technology-product descriptions, product features, target audience information, campaign objectives, geographical region, and available marketing budget. The collected data will be cleaned, anonymized where necessary, and prepared for analysis. Textual responses will undergo NLP preprocessing, while structured questionnaire responses will be encoded into suitable numerical or categorical representations for subsequent analysis and evaluation.

3.3 Baseline / Benchmark

The proposed Marketing Strategy Assessment will be evaluated against baseline approaches appropriate to each major function of the component. For audience technical-literacy classification, a rule-based classification approach derived from structured questionnaire scores will be considered as a baseline and compared with the proposed NLP-assisted approach. For feature-to-value translation, manually developed customer-oriented messages will be used as a baseline for comparison with AI-generated messages. For marketing-channel recommendation, a simple rule-based channel-selection approach will be considered as a baseline and compared with the proposed AHP-based ranking approach. For budget allocation, an equal-allocation strategy across selected channels will be considered as a baseline against the proposed budget allocation approach. Expert assessments will also be used where appropriate to provide an external basis for evaluating the relevance and quality of the generated outputs.

3.4 Evaluation Metrics

The proposed Marketing Strategy Assessment will be evaluated using separate metrics for its major functional components. For the audience technical-literacy classification module, classification accuracy, precision, recall, and F1-score will be used to measure how accurately the system assigns audiences to the defined technical literacy categories. For the feature-to-value translation module, the generated marketing messages will be evaluated through expert assessment based on criteria such as clarity, relevance, accuracy, customer orientation, and appropriate technical detail. For the marketing-channel recommendation module, recommendation relevance and expert agreement will be used to assess whether the recommended channels are appropriate for the given audience and campaign objectives. For the budget allocation module, the proposed allocation will be compared with the baseline allocation strategy using criteria such as alignment with channel suitability, objective achievement, and expert assessment. Overall system usability will be evaluated using a structured usability questionnaire, while qualitative feedback from relevant stakeholders will be used to identify strengths, limitations, and areas for improvement.

Metric	Why Relevant	Target / Comparison
Accuracy	Measures the proportion of target audiences correctly classified into the defined technical-literacy categories.	Compare the proposed NLP-assisted classification model with the rule-based baseline.
Precision, Recall and F1-score	Provide a more detailed evaluation of classification performance, particularly when the number of samples in each technical-literacy category is unequal.	Compare the proposed model with the rule-based baseline for Low, Medium, and High technical-literacy classes.
Expert Assessment Score	Measures the quality of AI-generated feature-to-value messages in terms of clarity, relevance, accuracy,	Compare AI-generated messages with manually prepared baseline messages using a structured rating scale.

Metric	Why Relevant	Target / Comparison
	customer orientation, and appropriate technical detail.	
Recommendation Relevance	Determines whether the recommended marketing channels are suitable for the target audience, campaign objective, and marketing context.	Compare the AHP-based recommendations with the rule-based channel-selection baseline and expert judgments.
Budget Allocation Deviation	Measures how closely the proposed allocation corresponds to expert-recommended or benchmark allocations across selected marketing channels.	Compare the proposed allocation with an equal-allocation baseline and expert-recommended allocation.
Usability Score	Measures whether intended users can understand and use the system outputs effectively for marketing decision-making.	Evaluate using a structured usability questionnaire such as the System Usability Scale (SUS), where applicable.

Table 2.1Evaluation Metrics for the Marketing Strategy Assessment

3.5 Validation Strategy

The proposed Marketing Strategy Assessment will be validated through a combination of quantitative testing and expert-based validation. The technical-literacy classification module will be validated by comparing its predicted classifications with reference classifications derived from questionnaire-based technical-literacy assessments. The feature-to-value translation module will be validated by obtaining evaluations from marketing or technology-domain experts based on the defined quality criteria. The marketing-channel recommendation and budget allocation modules will be validated by comparing their outputs with baseline approaches and expert recommendations. System usability will be validated by allowing representative users to interact with the prototype and complete a structured usability questionnaire. Feedback obtained during validation will be analyzed to identify errors, inconsistencies, and areas requiring improvement, and the system will be refined iteratively based on the findings.

Validation Plan

Component	Validation Method
Technical literacy classification	Compare predicted classes with reference classifications
Feature-to-value translation	Expert evaluation using predefined quality criteria
Channel recommendation	Compare AHP recommendations with baseline and expert recommendations

Component	Validation Method
Budget allocation	Compare proposed allocation with baseline and expert allocation
Overall usability	User testing + structured usability questionnaire

Table 3 Validation Plan for the Marketing Strategy Assessment

3.6 Development Approach

The Marketing Strategy Assessment will be developed using an iterative prototyping approach. The development process will be divided into several stages, beginning with requirements analysis and data preparation, followed by the development of the individual system modules. The audience technical-literacy module will be developed using structured questionnaire data and textual inputs with NLP-assisted analysis. The feature-to-value translation module will use AI techniques to generate customer-oriented marketing messages from technical product information. The marketing-channel recommendation module will implement an AHP-based multi-criteria ranking approach, while the budget allocation module will use the resulting channel recommendations and user-defined budget constraints to generate an allocation plan.

An initial prototype will be developed to integrate these modules into a single workflow. The prototype will then be tested using representative inputs and evaluated according to the defined evaluation metrics and validation strategy. Feedback and evaluation results will be used to identify errors and improve the system iteratively. The final version will integrate the validated modules and provide the outputs through a marketing strategy dashboard.

Development Stages

Stage	Main Activity	Output
1	Requirements analysis	Functional and non-functional requirements
2	Data preparation	Cleaned questionnaire and textual data
3	Module development	Classification, translation, recommendation and allocation modules
4	Prototype integration	Initial Smart Marketing System
5	Testing and evaluation	Performance and usability results
6	Refinement	Improved system based on validation findings
7	Final integration	Completed Smart Marketing System

Table 4 Development Stages of the Marketing Strategy Assessment

4. HIGH LEVEL SYSTEM ARCHITECTURE

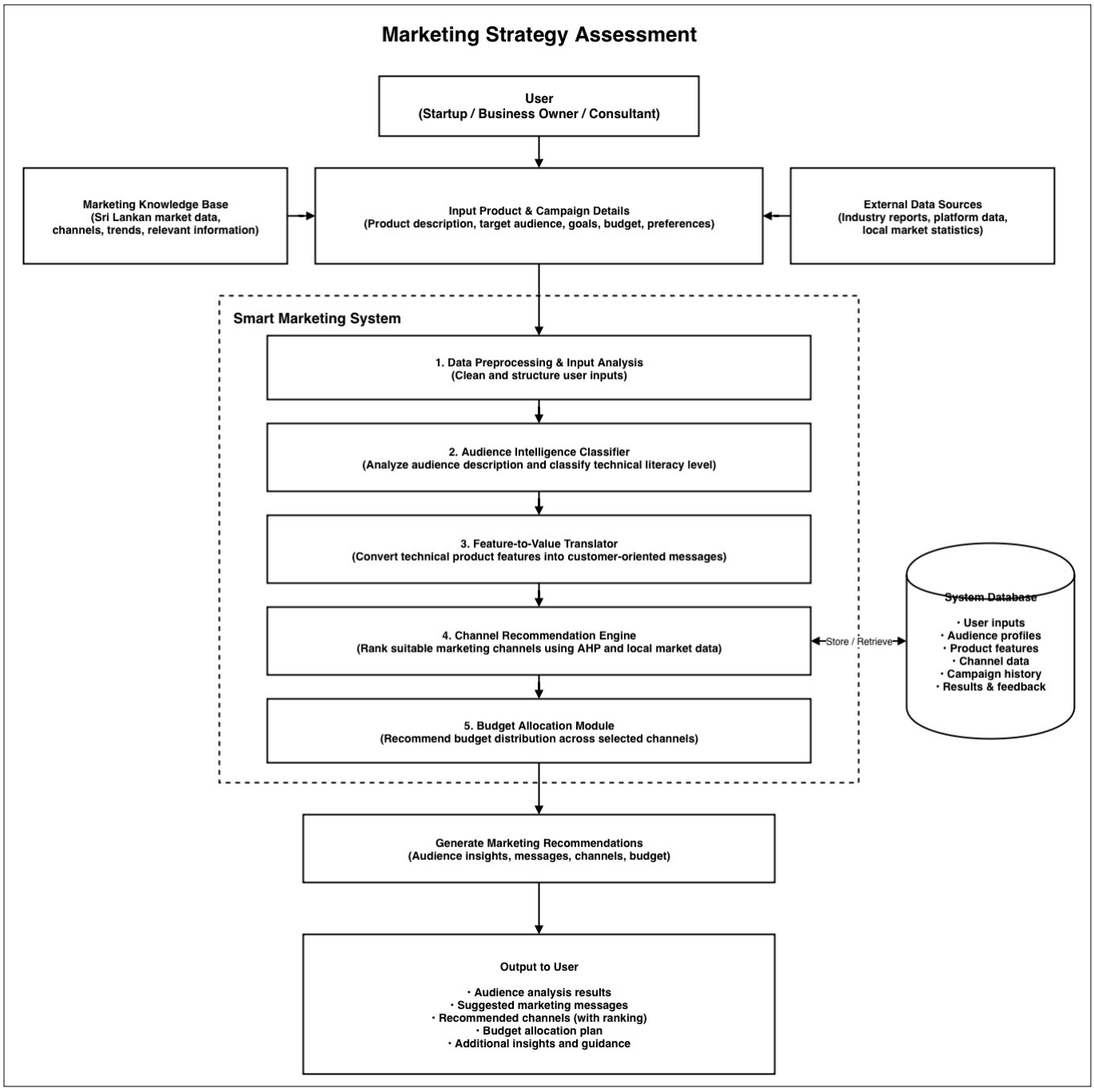


Figure 1High-Level Architecture of the Marketing Strategy Assessment

5. USER REQUIREMENTS

5.1 Functional Requirements

The preliminary functional requirements of the Marketing Strategy Assessment were derived from the identified research problem, literature review, proposed system objectives, and planned stakeholder requirements analysis. These requirements will be further validated and refined using questionnaire and interview findings during the research.

ID	Functional Requirement	Description	Verification
FR1	Product Information Input	The system shall allow users to enter product/service descriptions, technical features, and the problem solved by the product/service.	Enter sample product information and verify that it is accepted and processed.
FR2	Audience Information Input	The system shall allow users to provide target-audience characteristics, including customer type, technical familiarity, and geographical information.	Enter different audience profiles and verify successful processing.
FR3	Technical Literacy Analysis	The system shall analyze structured questionnaire responses and textual audience information to classify the target audience into defined technical-literacy categories.	Compare system classifications against reference classifications.
FR4	Feature-to-Value Translation	The system shall generate customer-oriented marketing messages from technical product features according to the characteristics of the target audience.	Evaluate generated messages using predefined expert-assessment criteria.
FR5	Marketing Channel Recommendation	The system shall rank suitable marketing channels based on audience characteristics, campaign objectives, and selected evaluation criteria using AHP.	Compare recommendations with the defined baseline and expert assessment.
FR6	Budget Allocation	The system shall generate a recommended allocation of a user-defined marketing budget among selected marketing channels.	Compare the proposed allocation against the selected baseline and expert recommendation.
FR7	Marketing Strategy Output	The system shall present the audience profile, generated messages, recommended channels, channel ranking, and budget allocation in an understandable format.	Verify that all required outputs are displayed for a valid input scenario.

Table 5 Functional Requirements of the Marketing Strategy Assessment

5.2 Non Functional Requirements

The Marketing Strategy Assessment should also satisfy non-functional requirements related to usability, performance, reliability, security, and maintainability. These requirements will be considered during prototype development and evaluated where applicable.

ID	Non-Functional Requirement	Measurable Criterion
NFR1	**Usability**	Users should be able to complete the main marketing-analysis workflow without technical assistance, and usability should be assessed using a structured usability questionnaire such as SUS.
NFR2	**Performance**	The system should return recommendations and generated outputs within an acceptable response time during prototype testing. The measured response time will be recorded during evaluation.
NFR3	**Accuracy**	The technical-literacy classification model should be evaluated using accuracy, precision, recall, and F1-score against the defined reference classifications.
NFR4	**Reliability**	The system should produce outputs consistently when the same valid input is processed repeatedly under the same conditions.
NFR5	**Security and Privacy**	Questionnaire and company information should be stored and processed with appropriate access controls, and personally identifiable information should be anonymized where required.
NFR6	**Maintainability**	The system should use modular components so that the classification, message-generation, channel-recommendation, and budget-allocation modules can be updated independently.

Table 6 Non-Functional Requirements of the Marketing Strategy Assessment

6. COMMERCIALIZATION PLAN

6.1 Target Market and User Persona

The primary target market for the Marketing Strategy Assessment is Sri Lankan technology companies, particularly technology start ups and small and medium sized enterprises (SMEs) that market software products, digital services, technology solutions, or technology enabled products. The primary users are founders, business owners, marketing managers, and marketing executives who are responsible for making marketing decisions but may have limited access to specialized marketing analytics and decision support resources.

A representative user persona is a Sri Lankan technology company founder or marketing manager who needs to promote a technology product to customers with different levels of technical knowledge while working within a limited marketing budget. The user requires support in understanding the target audience, communicating technical product features in a customer friendly manner, selecting appropriate marketing channels, and distributing the available marketing budget.

6.2 Value Proposition

The Marketing Strategy Assessment provides an AI assisted marketing decision support solution that brings several marketing decisions into a single workflow. Instead of requiring users to separately analyze their audience, prepare customer oriented messages, compare marketing channels, and plan budget allocation, the proposed system provides integrated recommendations based on product information, target audience characteristics, campaign objectives, and available budget. Its primary value is to help Sri Lankan technology companies make more structured and context-aware marketing decisions while reducing the complexity of converting technical product information into customer-oriented communication.

6.3 Revenue Model & Pricing

The proposed system can follow a Software-as-a-Service (SaaS) business model in which technology companies pay for access to the marketing decision support platform. A tiered subscription model can be considered, with different plans based on factors such as the number of marketing analyses, number of products or campaigns, and access to advanced recommendations. A free or limited trial version could also be provided to allow potential customers to evaluate the system before subscribing. The final pricing structure will be determined after considering development and operational costs, competitor offerings, and the affordability of the target Sri Lankan technology acompany market.

6.4 Competitive Advantage

The proposed Marketing Strategy Assessment can differentiate itself by integrating four marketing decision support functions within a single system: audience technical literacy analysis, AI assisted feature to value translation, multi criteria marketing channel recommendation, and marketing budget allocation. In addition, the system is designed specifically for the context of Sri Lankan technology companies and can incorporate relevant local marketing information into its recommendations. This combination provides a contextual advantage over general purpose marketing tools that primarily focus on individual campaign execution or isolated marketing functions.

6.5 Cost Consideration

The major operational costs of the proposed system are expected to include cloud infrastructure, database and storage services, AI API usage, vector database or embedding services where required, software development and maintenance, and system hosting. The cost structure will depend on the number of users, frequency of AI generated outputs, amount of stored marketing knowledge, and infrastructure requirements. These costs will be considered when developing the proposed pricing and commercialization model.

6.6 Intellectual Property

The intellectual property contribution of the proposed system will primarily consist of the software implementation, system architecture, processing workflow, datasets or structured research data developed during the project, and the integration of the proposed audience analysis, feature to value translation, channel recommendation, and budget allocation methods. The research will also consider the appropriate use and licensing of external datasets, APIs, libraries, and third party services used during development.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget

Item	Purpose	Estimated Cost (LKR)
Existing Laptop / Computer	Development, testing and documentation	0
NLP / Python Libraries	Text processing, classification and experimentation	0
AHP / AHPy Library	Marketing-channel ranking	0
Supabase	Database and application data storage	0
Vector Database / pgvector	Storage and retrieval of marketing knowledge embeddings	0
Gemini API	AI-assisted feature-to-value translation and marketing analysis	10,000
Web Application Hosting	Hosting the research prototype	5,000
Domain / Deployment Costs	Optional public prototype deployment	4,000
Data Collection / Printing	Questionnaires and stakeholder data collection	3,000
Contingency	Unexpected development and service costs	4,000
Estimated Total		**26,000**

Table 7 Estimated Budget for the Marketing Strategy Assessment

7.2 Budget Justification

Most software libraries required for the Marketing Strategy Assessment are open source and therefore do not require direct licensing costs. The main variable cost is expected to be AI API usage because the feature to value translation module may require external generative AI services during development and testing. Database, vector storage, and hosting costs may also remain within free or low cost tiers during prototype development, depending on service limitations and usage. Data collection costs include basic expenses associated with distributing questionnaires or conducting stakeholder activities. A contingency allocation is included to accommodate unexpected API, hosting, or development-related expenses.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure for the Marketing Strategy Assessment is derived from the team's Jira project timeline. The marketing component is divided into five main work packages covering stakeholder data collection, marketing input design, audience intelligence analysis, feature to value translation, and development of a Sri Lankan digital-market knowledge base and channel-related functionality. These activities collectively support the development and evaluation of the Smart Marketing System and are aligned with the component's research objectives and proposed architecture.

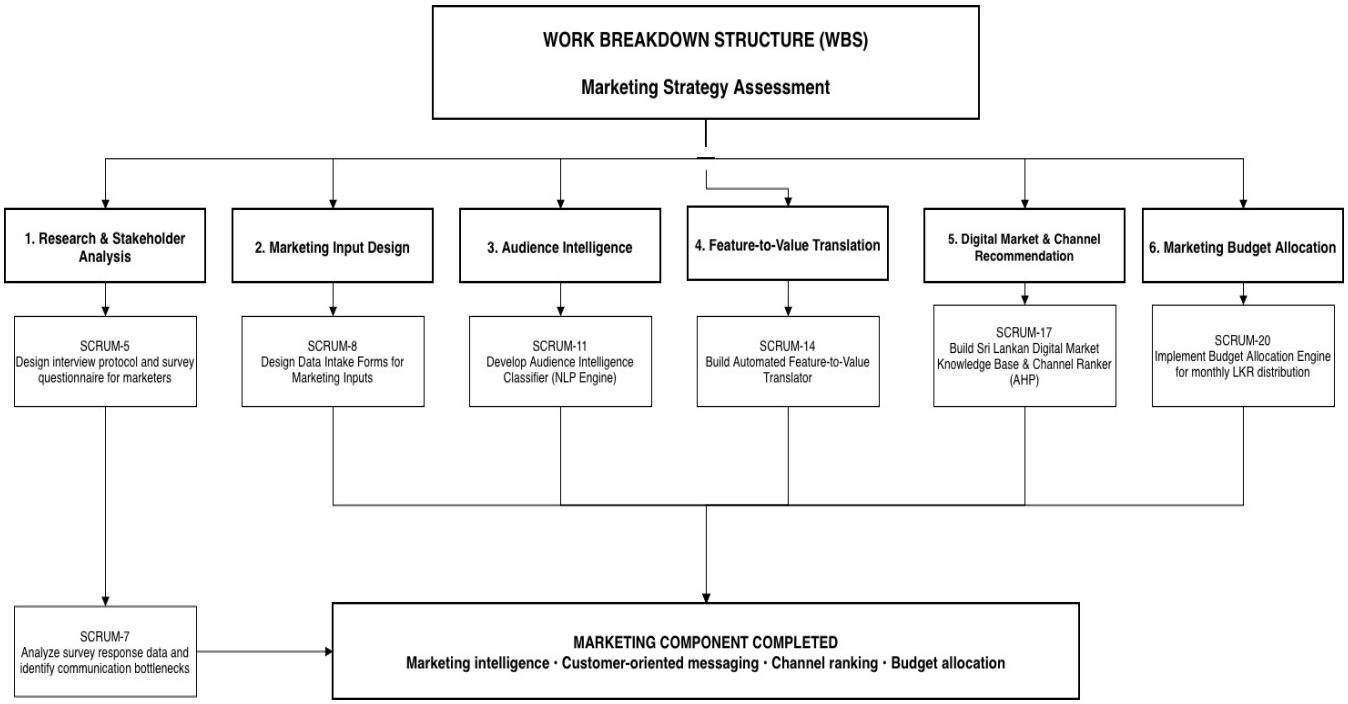


Figure 2 Work Breakdown Structure of the Marketing Strategy Assessment

9. GANTT CHART

The project timeline for the Marketing Strategy Assessment is derived from the team's Jira project schedule. The marketing component is planned over a 40 day period from 2 September 2026 to 11 October 2026. The activities are organized sequentially, beginning with stakeholder data collection and marketing-input design, followed by audience intelligence development, feature to value translation, and development of the Sri Lankan digital-marke

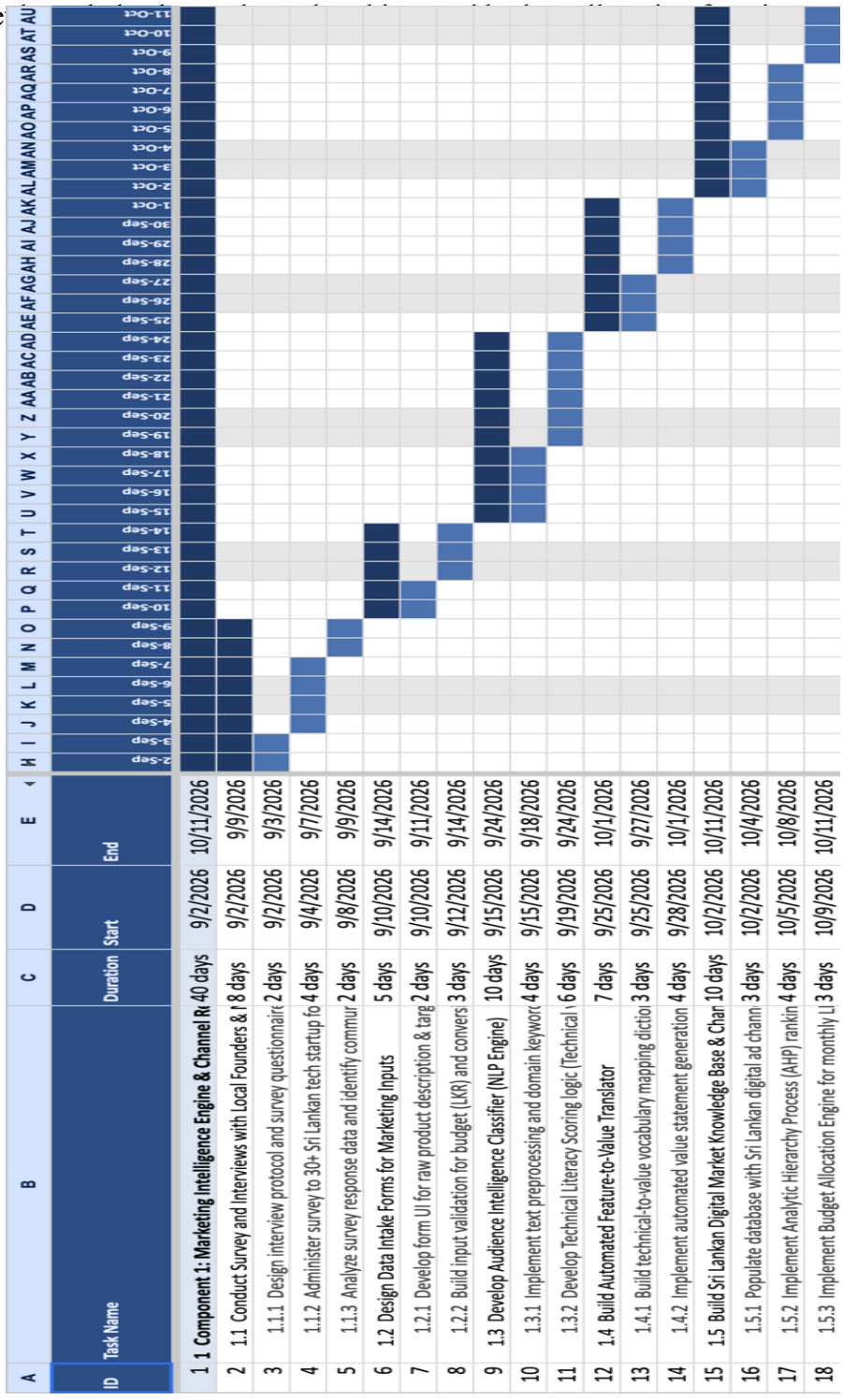


Figure 3 Gantt Chart of the Marketing Strategy Assessment

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11. APPENDICES

Appendix 1 – Stakeholder Questionnaire

Section A — Product Information

Q1. What is the name of your product or service?

Q2. Briefly describe your product or service.

Q3. What are the main features of your product?

Q4. What industry does your product belong to?

Q5. Who is your primary target customer?

Section B — Target Customer Information

Q6. Is your target market B2B, B2C, or both?

Q7. What age group best represents your target customer?

Q8. What is the typical occupation or role of your target customer?

Q9. Where is your target audience located?

Section C — Technical Literacy

Q10. How would you describe your target customer's comfort with technology?

Q11. How familiar is your target customer with technical terms related to your product?

Q12. How detailed should your marketing communication be for your target audience?

Section D — Customer Value and Marketing Channels

Q13. What is the main benefit you want customers to understand about your product?

Q14. What are the main customer pain points your product solves?

Q15. Which marketing channels are you currently using?

Q16. Which marketing channels have worked best for you so far?

Q17. What type of content does your target audience engage with most?

Section E — Marketing Budget and Campaign

Q18. What is your monthly marketing budget?

Q19. How flexible is your marketing budget?

Q20. What is your expected campaign duration?

Q21. What would define a successful campaign for your business?

Appendix 2 – Literature Comparison

Ref.	Research Area	Method / Approach	Context	Main Contribution	Limitation
[1]	Digital marketing adoption	TOE-based study	Sri Lankan SMEs	Identifies factors influencing digital marketing adoption	Does not provide an integrated AI marketing decision-support system
[2]	Technology startup challenges	Phenomenological study	Sri Lankan technology startups	Identifies startup-related challenges	Does not specifically address marketing decision support
[3]	Digital competence	Competence framework	General digital competence	Provides conceptual basis for digital competence	Not designed for marketing technical-literacy classification
[4]	NLP/text classification	ML/deep-learning review	Text-classification research	Provides technical foundation for text classification	Not specific to audience technical literacy in technology marketing
[5]	Customer value	Value Proposition Canvas	Product/customer value	Connects customer needs with product value	Does not automate feature-to-value translation
[6]	Channel selection	AHP	Online advertising	Demonstrates multi-criteria channel selection	Not tailored to Sri Lankan technology companies
[7]	Budget allocation	Nonlinear optimization	Digital media channels	Provides structured advertising budget allocation	Does not integrate technical literacy/local context
[8]	Digital market	Digital statistics	Sri Lanka	Provides local digital-market context	Descriptive rather than decision-support

Appendix 3 – Requirements and Objective Mapping

Objective	Related Requirement	System Function	Evaluation
O1	FR1, FR2	Marketing/product/audience input	Requirements analysis
O2	FR3, NFR3	Audience technical-literacy classifier	Accuracy, Precision, Recall, F1
O3	FR4	Feature-to-value translator	Expert assessment
O4	FR5	AHP channel recommendation	Recommendation relevance/expert comparison
O5	FR6, FR7, NFR1	Budget allocation and marketing output	Baseline comparison/usability

Appendix 4 – AI Use Disclosure and Evidence/Decision Log

Activity	AI Used?	Purpose	Human Verification / Decision
Literature search	Yes	Identify potentially relevant sources	Sources manually verified
Proposal drafting	Yes	Language and structure assistance	Content reviewed and edited
Architecture design	No	Initial design assistance	Final architecture manually designed/edited
WBS design	No	Structure generation	Jira tasks manually verified
Questionnaire design	Yes	Question drafting	Questions reviewed against research objectives
AI feature-to-value module	Yes	Proposed system functionality	Outputs evaluated using defined criteria